
IK 548 : Advanced NLP Techniques for Indian Languages

Course Code : IK 548

Course Name : Advanced NLP Techniques for Indian Languages

L-T-P-C : 3-0-0-3

Intended for : BTech/MTech/MS/MSc/MA/Ph.D

Prerequisite :

Mutual Exclusion: None

Approval: 42nd Senate

Course Contents

- **Natural Language Processing with Classification and Vector Spaces**

- Sentiment Analysis with logistic Regression (Total = 6 Hours)
 - * 15 videos - Total 84 minutes
 - * 13 readings - Total 100 minutes ● 1 quiz - Total 30 minutes
- Sentiment Analysis with Naive Bayes (6 Hours)
 - * 13 videos - Total 44 minutes
 - * 12 readings - Total 111 minutes
 - * 1 quiz - Total 30 minutes
- Vector Space Models (5 hours)
 - * 10 videos - Total 28 minutes
 - * 10 readings - Total 91 minutes
 - * 1 quiz - Total 30 minutes
- Machine Translation And Document Search (6 Hours)
 - * 11 videos - Total 68 minutes
 - * 10 readings - Total 91 minutes
 - * 1 quiz - Total 30 minutes

- **Natural Language Processing with Probabilistic Models**

- Autocorrect (4 hours)
 - * 11 videos - Total 31 minutes
 - * 10 readings - Total 37 minutes
 - * 1 quiz - Total 30 minutes
- Part Of Speech Tagging and Hidden Markov Models (5 hours)
 - * 13 videos - Total 42 minutes
 - * 12 readings - Total 66 minutes
 - * 1 quiz - Total 30 minutes
- Autocomplete and Language Models (5 hours)
 - * 11 videos - Total 53 minutes
 - * 10 readings - Total 70 minutes

- * 1 quiz - Total 30 minutes
- Word embedding with neural networks (8 hours)
 - * 22 videos - Total 73 minutes
 - * 22 readings - Total 88 minutes
 - * 1 quiz - Total 30 minutes
- **Natural Language Processing with Sequence Models**
 - Neural Networks For Sentiment Analysis (6 hours)
 - * 11 videos - Total 39 minutes
 - * 10 readings - Total 51 minutes
 - * 1 quiz - Total 30 minutes
 - Recurrent Neural Networks For Language Modeling (6 hours)
 - * 10 videos - Total 28 minutes
 - * 9 readings - Total 44 minutes
 - * 1 quiz - Total 30 minutes
 - LSTMs and Named Entity Recognition (5 hours)
 - * 8 videos - Total 25 minutes
 - * 10 readings - Total 53 minutes
 - * 1 quiz - Total 30 minutes
 - Siamese Networks (6 hours)
 - * 10 videos - Total 35 minutes
 - * 10 readings - Total 50 minutes
 - * 1 quiz - Total 30 minutes
- **Natural Language Processing with Attention Models**
 - Neural Machine Translation (7 hours)
 - * 15 videos - Total 88 minutes
 - * 6 readings - Total 37 minutes
 - * 1 quiz - Total 30 minutes
 - Text Summarization (5 hours)
 - * 10 videos - Total 39 minutes
 - * 8 readings - Total 62 minutes
 - * 1 quiz - Total 30 minutes
 - Question Answering (11 hours)
 - * 16 videos - Total 60 minutes
 - * 16 readings - Total 242 minutes
 - * 1 quiz - Total 30 minutes
 - Chatbot (6 hours)
 - * 9 videos - Total 63 minutes
 - * 13 readings - Total 141 minutes
 - * 1 quiz - Total 30 minutes

Textbooks:

1. Sue Knight, **NLP at Work**
2. Shelle Rose Charvet, **Words that change minds.**
3. Robert Dilts, **Sleight of mouth.**
4. Lynne Cooper, **Business NLP for dummies.**
5. **Mind line**

References:

1. NA